

Oahu Flood Model

vs. Insurance History

Storm-surge predictions compared to
45 years of NFIP claims (1978-2023)

kappa_spatial

0.352

Model-vs-claims spatial agreement

Moran's I

-0.358

Errors are spatially random, not clustered

Verdict

FAIL

Too weak to confirm geographic bias

Buildings

47,983

1,676 flooded by model (3.5%)

Predicted Loss

\$55M

\$33M building + \$22M contents

NFIP Claims

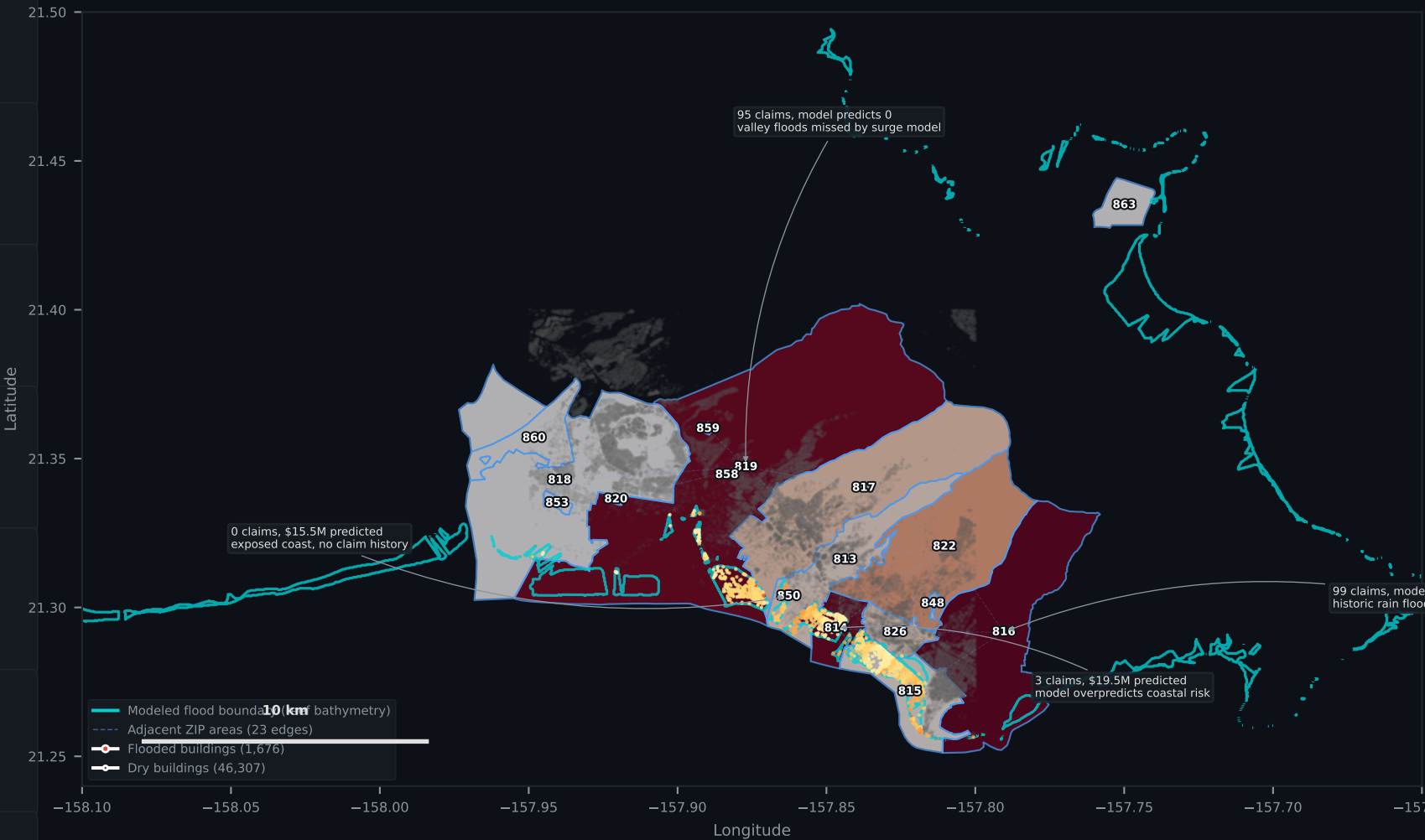
379

Federal flood insurance, 1978-2023

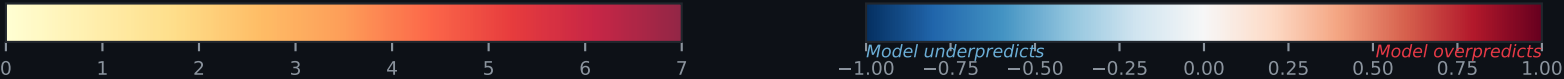
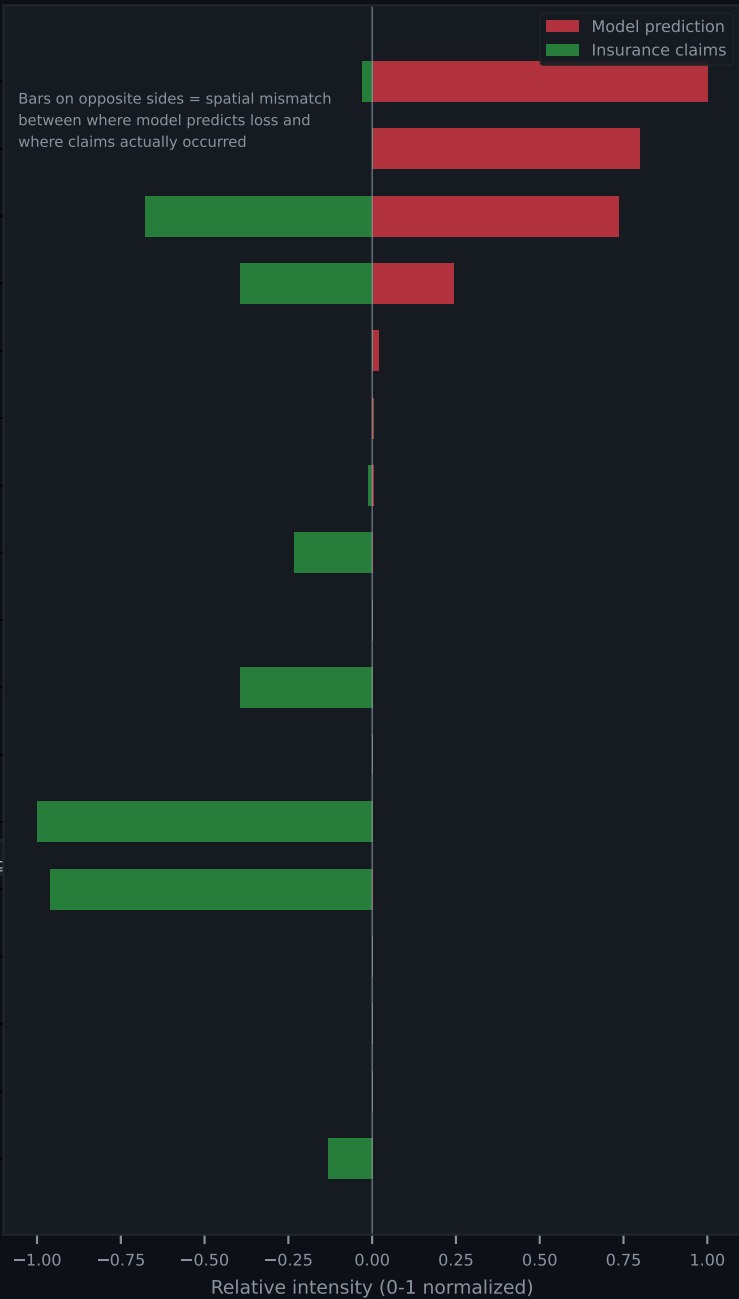
ZIP Areas

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Oahu ZCTAs with 968xx prefix



Where Model and Claims Disagree



NFIP = National Flood Insurance Program. Claims span 1978-2023 and are cumulative across all events, not matched to a single storm. Comparison is illustrative, not adjudicative.